

# Logistic Regression

(1)

LR: a typical case of binary classification

input features  $x = x_1, x_2, \dots, x_n$

binary output  $y \in \{0, 1\}$

we want to have a model that

spam  $\Rightarrow y=1$

Not spam  $\Rightarrow y=0$

Gives with :-

$$P(y=1|x)$$

saying that the class is 1 for given input  $x$  more precisely lets call it probability.

$$p(x) = P(y=1|x)$$

The probability is a function of " $x$ " giving output  $y \in \{0, 1\}$

$$P(y=0|x) = 1 - P(y=1|x) = 1 - p(x)$$

lets say that in order to predict the probability as function of " $x$ " we use linear expression.

$n \rightarrow \# \text{ of features}$

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

$$z = \beta^T x$$

$z$  is called linear predictor or score.

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}, x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$
$$\beta^T = [\beta_0 \ \beta_1 \ \beta_2 \ \dots \ \beta_n]$$

We can't directly represent  $p(x) = z$  because

mismatch  $\swarrow$

$$\begin{array}{l} -\infty < z < +\infty \\ \uparrow \\ 0 \leq p(x) \leq 1 \end{array}$$

So the linear model can't be used directly as linear predictor for the probability function  $p(x)$  because we may get impossible values like 3.9 & -10.20 as " $z$ " is a real value function.

Solution is only possible if by some how we can bring the range  $-\infty < z < +\infty$  in equivalent range of  $0 \leq p(x) \leq 1$

Odds : lets say  $p$  is the probability for given  $x$  when  $y=1$   
 $p = P(Y=1|x)$

then the prob when  $y=0$  for that same  $x$  will be  $1-p$

The odds of class 1 is

$$\text{odds}_1 = \frac{p}{1-p}$$

if  $p=0.5$ , odds=1

odds range 0 to  $\infty$

if  $p > 0.5$ , odds  $> 1$

if  $p < 0.5$ , odds  $< 1$

take log of odds (called log-odds or logit)

$$\text{log-odds}_1 = \log\left(\frac{p}{1-p}\right)$$

↳ result in range  $(-\infty, \infty)$

Now, instead of  $p(x) = x$  we can write

$$\left[ \log\left(\frac{p}{1-p}\right) = x \right] \rightarrow (I)$$

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

take exponential of  $(I)$

$$e^{\log(p/(1-p))} = e^x$$

$$e^{\log(a)} = a$$

$$\frac{p}{1-p} = e^x \Rightarrow p = e^x(1-p)$$

$$p = e^x - e^x p \Rightarrow p + e^x p = e^x \Rightarrow p(1 + e^x) = e^x$$

$$p = \frac{e^x}{1 + e^x} \Rightarrow p = \frac{e^x/e^x}{(1 + e^x)/e^x} = \frac{1}{\frac{1}{e^x} + e^x/e^x}$$

$$p = \frac{1}{1 + e^{-x}}, \text{ giving name}$$

$$p = \sigma(x) = \frac{1}{1 + e^{-x}}$$

log-odds are linear



So

$$P(Y=1|x) = \frac{1}{1 + e^{-x}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n)}}$$

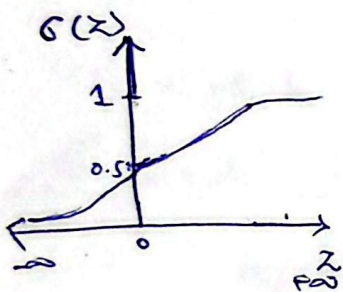
$$P(Y=0|x) = 1 - P(Y=1|x)$$

if  $x \rightarrow +\infty$ 

$$e^{-x} \rightarrow 0 \quad \left[ e^{-x} = \frac{1}{e^x} = \frac{1}{e^{+\infty}} = \frac{1}{\text{very large number}} \approx 0 \right]$$

if  $x \rightarrow -\infty$ 

$$e^{-x} \rightarrow +\infty \quad \left[ e^{-x} = \frac{1}{e^x} = \frac{1}{e^{-\infty}} = e^{+\infty} \approx \text{very large number} \right]$$



- So if  $x \rightarrow +\infty$  then  $e^{-x} \rightarrow 0$  &

hence  $\sigma(z) \rightarrow 1$

- if  $x \rightarrow -\infty$  then  $e^{-x} \rightarrow +\infty$  &

hence  $\sigma(z) \rightarrow 0$

- if  $x = 0$

$$\sigma(z) = 1/2$$

positive score ( $z > 0$ ) mean probability above 0.5

negative score ( $z < 0$ ) means probability below 0.5

Zero score mean probability exactly 0.5

Thus for an input instance  $x_i$ , logistic regression predicts a probability

$$p_i = P(Y_i=1|x_i)$$

$$1 - p_i = P(Y_i=0|x_i)$$

$$P(Y_i|x_i) = p_i^{y_i} (1 - p_i)^{1 - y_i}$$

Case 1:  $y_i = 1$   
 $P(Y_i=1|x_i) = p_i^1 (1 - p_i)^{1-1} = p_i$

Case 0

$$p_i^0 (1 - p_i)^{1-0} = 1 - p_i$$

For  $m$  training examples

$$(x_1, y_1), (x_2, y_2), (x_3, y_3) \dots (x_m, y_m)$$

Assume observations are independent. Prob of seeing the whole dataset given parameter  $\beta$  is simply the product of individual probabilities

$$L(\beta) = \prod_{i=1}^m P(y_i | x_i)$$

or

$$L(\beta) = \prod_{i=1}^m p^{y_i} (1-p)^{1-y_i} \rightarrow \text{called likelihood function}$$

We want to choose  $\beta$  so that observed data becomes as likely as possible. So

we solve  $\beta' = \arg \max_{\beta} L(\beta) \rightarrow \text{maximum likelihood}$

to convert products to sums we take log

$$\ell(\beta) = \log L(\beta)$$

$$\log(ab) = \log a + \log b$$

$$\log a^b = b \log a$$

$$= \log \left( \prod_{i=1}^m p^{y_i} (1-p)^{1-y_i} \right)$$

$$\ell(\beta) = \sum_{i=1}^m [y_i \log p + (1-y_i) \log (1-p)]$$

$\hookrightarrow$  log likelihood

Instead of maximizing the log likelihood, we minimize the negative log-likelihood

$$J(\beta) = -\ell(\beta)$$

also note

$$\frac{d}{dx} \sigma(x) = \sigma(x) * (1 - \sigma(x))$$

$$J(\beta) = - \sum_{i=1}^m [y_i \log p + (1-y_i) \log (1-p)]$$

called log loss or Binary cross entropy